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Communication

Copper-Catalyzed Asymmetric Alkynylation of sp^2 Allenic C−H Bonds for the Synthesis of Skipped Diynes

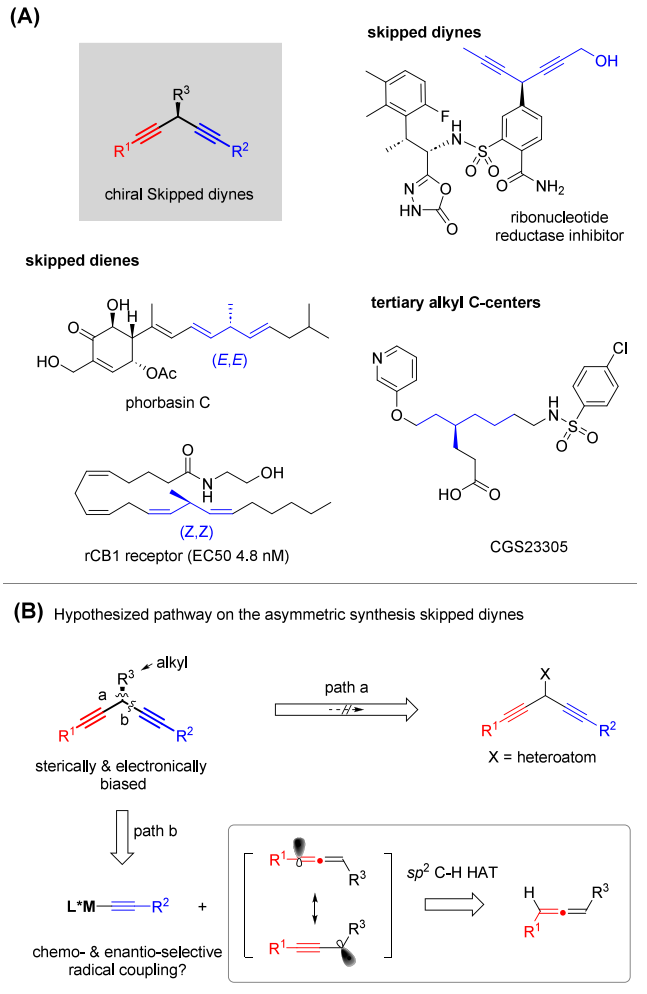
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Cite This: J. Am. Chem. Soc. 2025, 147, 46847−46853
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ABSTRACT: Herein, we established a copper-catalyzed asymmetric alkynylation of the sp^2 C−H bond of silylallenes, which was presented as an efficient method for the synthesis of enantioenriched skipped diynes with excellent enantioselectivity. The reaction was initiated by hydrogen atom abstraction of sp^2 allenic C−H bonds, in which the resonance forms of allenic and propargylic radicals were chemo- and enantioselectively trapped by chiral alkynyl-Cu(II) species to deliver a variety of skipped diynes. The method features a broad substrate scope and robust functional group tolerance, accommodating both alkynyl silanes and boronic esters. Moreover, the resulting skipped diynes serve as versatile key synthons for incorporating enantiomerically enriched enynes, alkynes, alkanes, and their derivatives, which could be applied in the concise synthesis of several bioactive molecules.

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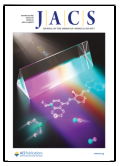
Scheme 1. Synthesis of Chiral Skipped Diynes



Skipped diynes—structures characterized by two alkyne groups bonded to a single carbon atom—are recognized as a key moiety in medicinal discovery and organic synthesis.[1] For instance, skipped diyne architectures have been positioned as key pharmacophoric elements in rational drug design.[2] Meanwhile, skipped diynes also serves as a pivotal synthon for the synthesis of skipped dienes via selective reduction, which extensively exists in natural products and bioactive compounds,[3] such as phorbasin C[4] and the arachidonoyl ethanolamide (AEA) analogue exhibiting exceptional rCB1 receptor affinity.[5] Moreover, full reduction of chiral skipped diynes presents an effective approach for the construction of enantiomerically enriched tertiary alkyl C-centers—structural units extensively pervasive in natural products and pharmaceuticals, such as sex pheromones,[6] methanophenazine,[7] and thromboxane synthase inhibitor CGS23005.[8] Therefore, exploiting efficient methods for the synthesis of optically pure skipped diynes is highly demanded.

Over the past decades, substantial efforts have led to numerous methods for synthesizing functionalized skipped diynes,[9] including those based on Lewis acid,[10] metal catalysis,[11] and others.[12] Despite these advances, their asymmetric synthesis remains a significant challenge. For example, nucleophilic substitution of skipped diynes is challenged due to the steric and electronic bias of the two alkyne moieties (path a, Scheme 1B). Alternatively, coupling two different alkynes offers a promising pathway, exemplified by the asymmetric coupling of propargylic radicals with alkynyl−metal species (pathway b).[13] In this context, generating propargylic radicals via hydrogen atom transfer (HAT) is particularly attractive, as it enables

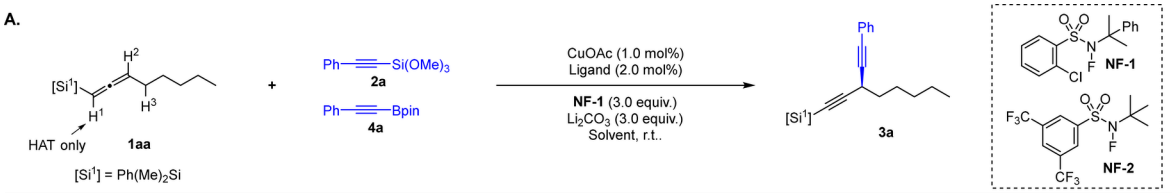
Received: October 31, 2025
Revised: December 1, 2025
Accepted: December 3, 2025
Published: December 8, 2025



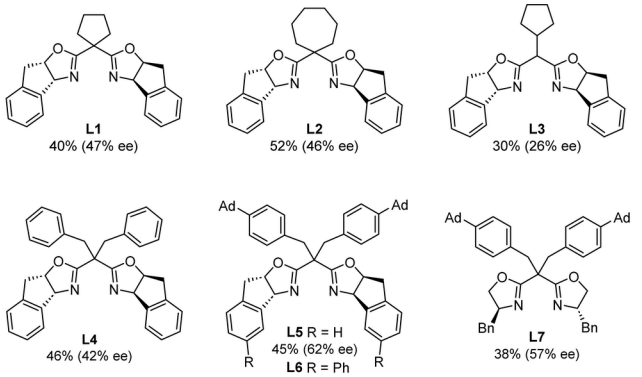
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https://doi.org/10.1021/jacs.5c19257
J. Am. Chem. Soc. 2025, 147, 46847−46853

Scheme 2. A. Site- and Enantioselective Allenic sp^2 C−H Alkynylation Reaction of 1aa with Alkynylation Reagent 2a and 4a: (*i*) Ligand Screening; (*ii*) Further Optimizing Reaction Conditions; B. The Reaction of 1ba with Alkyl-Substituted Alkynyl Reagents 2b and 4b



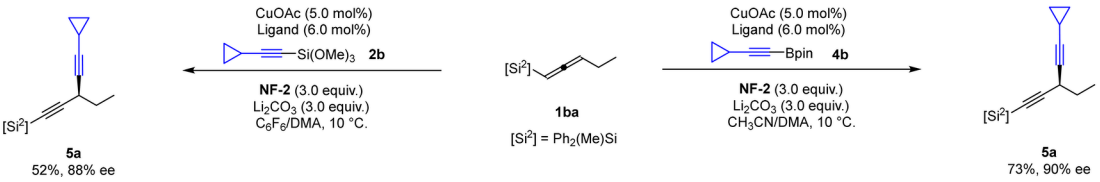
i) Ligand screening.[a]



ii) Optimizing reaction conditions.[a]

Entry	Ligand	Substrate	Solvent	Yield	*ee*
1	**L5**	**2a**	C_6F_6/DMAc	45%	62%
2	**L5**	**2a**	DCM/DMAc	25%	38%
3	**L5**	**2a**	Acetone/DMAc	45%	18%
4	**L6**	**2a**	C_6F_6/DMAc	39%	96%
5[b]	**L6**	**2a**	C_6F_6/DMAc	56%	96%
6[b,c]	**L6**	**2a**	C_6F_6/DMAc	50%	97%
7[b,c]	**L6**	**2a**	C_6F_6	51%	97%
8[**b,c,d**]	**L6**	**2a**	C_6F_6	**77%**	**97%**
9[b,c,d]	**L6**	**4a**	C_6F_6	58%	97%
10[b,c,d]	**L6**	**4a**	C_6F_6/DMAc	74%	97%
11[**b,c,d**]	**L6**	**4a**	MeCN/DMAc	**76%**	**97%**

B.



[a]Reaction conditions: CuOAc (1 mol %), **L6** (2 mol %), **1aa** (0.2 mmol), **2a** (0.6 mmol), **NF-1** (0.6 mmol), Li_2CO_3 (0.6 mmol), C_6F_6 (0.4 mL), rt, under Ar for 96 h. The crude ^{1}H NMR yields were determined by using TCE as an internal standard, and ee values were determined by HPLC on a chiral stationary phase. [b]**NF-1** was replaced **NF-2**. [c]10 °C. [d]CuOAc (5.0%), **L6** (6.0%).

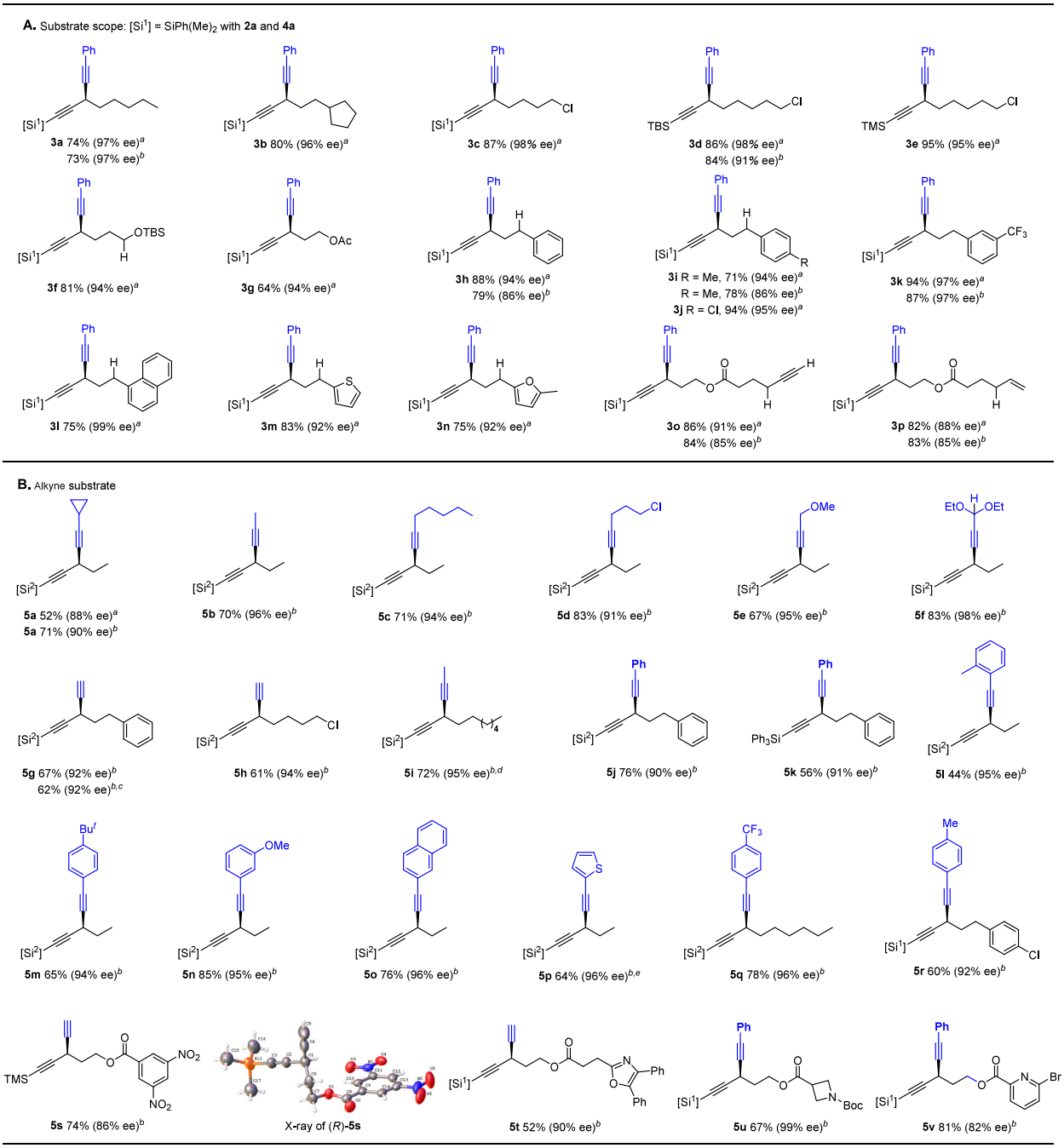
asymmetric alkynylation of C−H bonds. Herein, we report a copper-catalyzed asymmetric alkynylation of allenes for the synthesis of enantioenriched skipped diynes, in which the coupling partner of the resonance forms of allenic and propargylic radicals[14] is derived from abstraction of sp^2 allenic C−H bonds (Scheme 1B).

Building on our ongoing interest in asymmetric C−H functionalization, we have developed a copper-catalyzed radical relay strategy for this goal.[15] Recently, we disclosed an enantioselective C−H cyanation of allenes, in which site-selective HAT of the allenic sp^2 C−H bond of 1,3-allenes was achieved.[16a] In these studies, the resulting resonance forms of allenic radical and propargylic radical are quenched by a chiral Cu(II) cyanide species, where the coupling process often exclusively occurred at the allenic radical position to deliver allene products.[16−18] With a similar pathway, site-selective alkynylation of sp^2 C−H bond allenes has been established as well.[16b] Therefore, we hypothesized that, if this coupling process could be switched to the propargylic radical position in an enantioselective manner, an asymmetric C−H alkynylation of allenes would provide access to enantiomerically enriched skipped diynes.

Inspired by our recent study,[16d] our initial investigation focused on the reaction of silyl-substituted allene **1aa** and trimethoxysilyl-alkyne **2a**, using **NF-1** as an N-radical precursor and Li_2CO_3 as a base in perfluorobenzene/DMA (5:1 v/v). Pleasingly, a catalyst system involving CuOAc and Box ligands enabled allenic sp^2 C−H alkynylation for the synthesis of the skipped diynes **3a** with excellent chemoselectivity, while completely suppressing the formation of other coupling products (e.g., alkynyl-allene) derived from the allenic radical.[16b] More importantly, excellent site-selective HAT exclusively occurred at the allenic sp^2 C−H^1 bond, contributed by the fact that introducing a silyl group could significantly weaken the allenic sp^2 C−H bond.[19] Of the ligands evaluated, Box **L5** proved to be the most effective, delivering product **3a** in 45% yield and 62% ee (Scheme 2A-*i*). Further solvent screening indicated that replacing perfluorobenzene with solvents such as dichloromethane or acetone failed to improve the yield and led to a substantial erosion in enantioselectivity (entries 1−3, Scheme 2A-*ii*). Therefore, further modification of ligand Box **L5** by introducing a phenyl group into the oxazoline ring markedly increased the enantioselectivity to 96% ee but with a decreased yield (39%, entry 4). Subsequently, evaluation of oxidants revealed that the reaction with **NF-2** could increase the yield to 56% with 96% ee (entry 5). Moreover, when the temperature was lowered to 10 °C, a slight increase in enantioselectivity (97% ee) and a slight decrease in yield (50%) were observed, accompanied by an improved mass balance (entry 6). Distinct from our previous study,[20a,b] the removal of cosolvent DMAc gave a similar result (entry 7). Finally, the slightly low yield could be compensated by increasing the catalyst loading to 5 mol %,

https://doi.org/10.1021/jacs.5c19257
J. Am. Chem. Soc. 2025, 147, 46847−46853

Scheme 3. Substrate Scope on the Asymmetric Alkynylation of sp^2 C−H of Allenes; the Reaction Was Conducted on a 0.2 mmol Scale



[a]The reaction using silyl-substituted alkynyl reagent **2** (conditions see Scheme 2A-*ii*, entry 8). [b]The reaction using boronic ester-substituted alkynyl reagent **4** (conditions see Scheme 2B, entry 11). [c]5 mmol scale. [d]0.5 mmol scale. [e]at 0 °C.

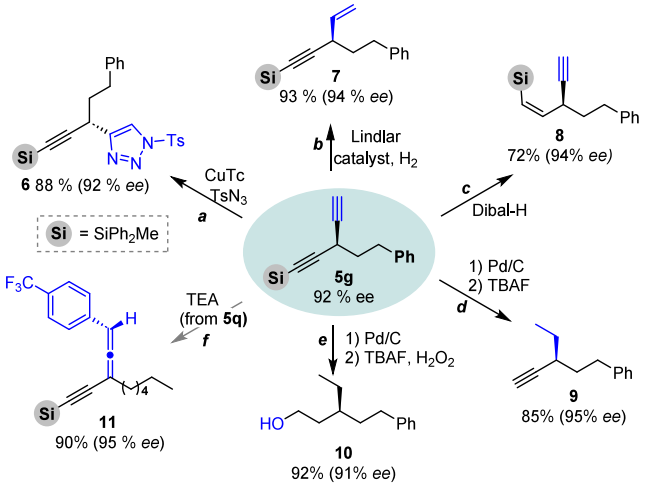
affording the best results in 77% yield with 97% ee (entry 8). Encouraged by the study of Kramer,[20c] we also investigated the reactivity of the alternative alkynyl boronate ester **4a**, demonstrating that it was also effective for the asymmetric C−H alkynylation. In this case, the addition of DMAc as a cosolvent is beneficial for a higher yield. A similar result (76% yield with 97% ee) was obtained in a mixture of CH_3CN and DMAc (entries 9−11).

Finally, alkyl-substituted alkynyl reagents **2b** and **4b** were also surveyed in the reaction of allene **1ba**, and these reactions proceeded smoothly under the above reaction conditions, affording the desired skipped diyne **5a** with excellent enantioselectivity (88−90% ee) (Scheme 2B). In this case, the reaction with alkynyl boronic ester **4b** provided better performance over the related silyl reagent **2b**.

With the optimized reaction conditions in hand, the substrate scope was investigated with diverse allenes and alkynyl reagent **2a** (Scheme 3A). Allene substrates bearing simple alkyl chains afforded the corresponding products **3a** and **3b** in both excellent yields (73−80%) and excellent enantioselectivities (96−97% ee). Moreover, various functional groups, such as halides (**3c**−**3e**), ether (**3f**), and ester (**3g**), were compatible to afford the

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Scheme 4. Synthetic Transformations ([Si] = $SiPh_2Me$)[a]



[a]Reaction conditions: (a) CuTc (10 mol %), $ArSO_2N_3$, in toluene, rt. (b) Lindlar's catalyst in EA, H_2, rt. (c) Dibal-H in THF, 70 °C. (d) (1) Pd/C in $CHCl_3$, H_2, rt. (2) TBAF in THF, rt. (e) (1) Pd/C in MeOH, H_2, rt. (2) TBAF, KF, $KHCO_3$, and H_2O_2. (f) TEA in THF, 0 °C.

corresponding products. Notably, allenes with other silyl groups, such as *tert*-butyldimethylsilyl (TBS, **3d**) and trimethylsilyl (TMS, **3e**), were compatible with the asymmetric C−H alkynylation conditions, affording the desired products with excellent yields (>85%) and enantioselectivities (>95% ee). More importantly, when allene substrates contained other active sp^3 C−H bonds, such as benzylic (**3h**−**3n**), propargylic (**3o**), and allylic (**3p**) C−H bonds, the HAT reaction still exclusively occurred at the allenic sp^2 C−H position, affording the related products with excellent enantioselectivity (88−99% ee). Particularly, heteroarenes, such as thiophene (**3m**) and furan (**3n**), were also tolerated under the standard conditions. In addition, alkynyl reagent **4a** also exhibited a broad substrate scope, affording a variety of skipped diynes (**3a**, **3d**, **3h**, **3i**, **3k**, **3o**, and **3p**) in comparable yields and enantioselectivities.

We turned our attention to investigating various alkyl-substituted alkyne reagents for asymmetric allenic sp^2 C−H akynylation. The results in Scheme 3B indicated that alkynyl silyl reagent **2a** was less effective than its boronic ester counterpart **4a**. Considering the superior performance, coupled with their greater stability and ease of synthesis, alkynyl boronic esters were chosen as the superior substrate class for subsequent scope investigation. Variants of alkyl-substituted boronic esters, such as a long-chain derivative (**5c**) and propargyl (**5b**, **5i**), were proven to be good candidates, affording the desired skipped diynes in good yields (70−72%) with excellent enantioselectivities (94−96% ee). Other functional groups in alkynyl reagents, such as halogen (**5d**) and ether (**5e**), were also well tolerated under the standard reaction conditions. Remarkably, the alkynyl reagent with ketal groups (**5f**) exhibited excellent performance, with an 83% yield and 98% ee, where the HAT of hydrogen at the active ketal moiety was not observed. More importantly, the reactions employing terminal alkynyl reagents (**5g**, **5h**, **5s**, **5t**) also proceeded efficiently, affording excellent results. Moreover, the process could be scaled up without erosion of efficiency or enantioselectivity. The desired products allow for further terminal alkynyl transformations.

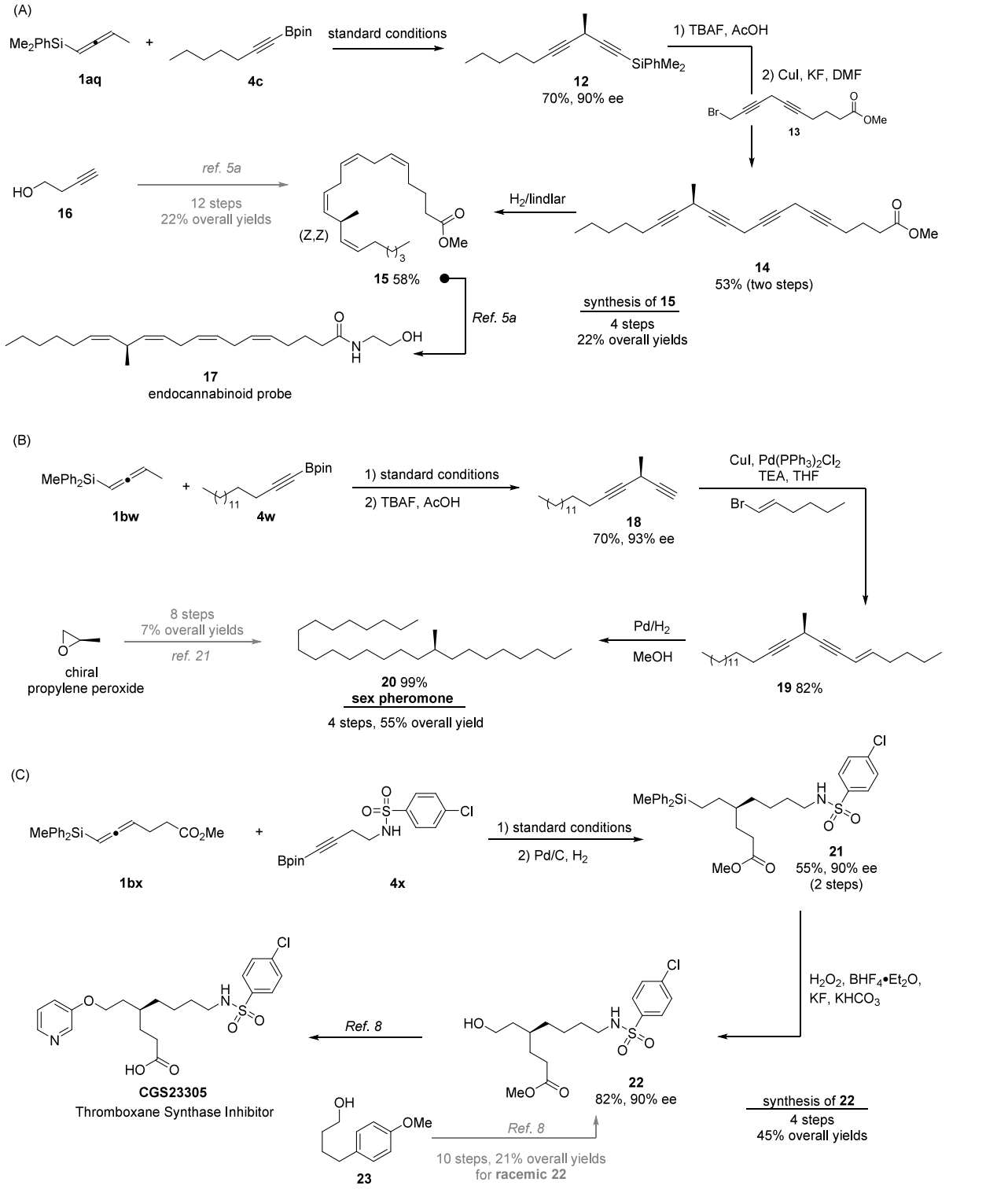
Moreover, we also examined various phenyl-substituted alkynyl boronic esters with different allenes. Compared to reagent **4a** with a phenyl group (**5j**, **5k**), the alkynyl reagents with various functionalized aryl groups provided a better performance on the enantioselectivity. For instance, reagents with aryl groups bearing ortho- and para-methyl (**5l**, **5r**), *tert*-butyl (**5m**), meta-methoxy (**5n**), naphthyl (**5o**), and trifluoromethyl (**5q**) afforded the corresponding skipped diynes in good yields with excellent enantioselectivities (92−96%). Meanwhile, a thiophene derivative (**5p**) was also compatible, providing product in 64% yield with 96% ee at 0 °C. For the allene substrates bearing heterocyclic (**5t**, **5v**) and high-strain rings (**5u**) the reactions performed well under standard conditions. Notably, the absolute stereoconfiguration of (*R*)-**5s** was confirmed by X-ray diffraction (XRD).

In general, skipped diyne products are recognized as versatile synthons in organic synthesis. As demonstrated in Scheme 4, the alkynylation product **5g** was readily converted to 1-sulfonyl-1,2,3-triazole **6** via the click reaction in 88% yield without enantiomeric erosion. In addition, site-selective semihydrogenation of **5g** either on acetylene or silyl-protected acetylene was achieved, affording enynes **7** in 93% yield (94% ee) using Lindlar's catalyst, or Z-**8** in 72% yield (94% ee) using diisobutylaluminum hydride (DIBAL-H), respectively. Moreover, the selective full hydrogenation of acetylene in **5g** was also conducted using Pd/C in $CHCl_3$, followed by desilylation to give chiral terminal alkyne **9** in 85% yield with 95% ee. By prolonging the reaction time with Pd/C in MeOH, both triple bonds in **5g** could be fully hydrogenated, followed by C−Si bond oxidation using hydrogen peroxide, affording the enantiomerically enriched alcohol **10** in 92% yield with 91% ee. Finally, the isomerization of **5q** is conducted by adding triethylamine, affording thermodynamically more stable allene **11** in 90% yield with 95% ee.

To demonstrate the synthetic utility of the method, as shown in Scheme 5, the formal synthesis of bioactive compounds or natural products was conducted using the current method as a key step. As an example, endocannabinoid probe **17**, an anandamide (AEA) analogue with exceptional CB1 receptor affinity, was synthesized using compound **15** as a key intermediate. The synthesis commenced with an asymmetric alkynylation of the sp^2 C−H bond in **1aq** with reagent **4c**, which provided skipped diyne **12**. Subsequent sequential desilylation and cross-coupling converted **13** into the corresponding tetrayne **14**. Finally, a selective semihydrogenation afforded the skipped tetraene, delivering probe **15** in 22% overall yield and 90% ee of 4 steps, which is much more efficient than the reported method from compound **16** with 12 steps, even in a similar overall yield.[5a] Second, the synthesis of sex pheromone **20** commenced from allene **1bw** and alkynyl reagent **4w**. Notably, this route required a streamlined, four-step sequence—asymmetric alkynylation, deprotection, Sonogashira coupling, and hydrogenation—to furnish the target molecule in 55% overall yield. This represents a substantial improvement in efficiency over the reported Zhong's method, which required eight steps from chiral propylene peroxide and delivered the product in only 7% overall yield.[21] Finally, for the chiral thromboxane synthase inhibitor CGS23005, its key intermediate **22** was concisely synthesized from allene **1bx** and alkynyl boronic ester **4x**, with the following reduction and oxidation. The 45% overall yield with 90% ee contrasts sharply with the reported 10-step procedure for the synthesis of racemic product in 21% overall yield.[8]

https://doi.org/10.1021/jacs.5c19257
J. Am. Chem. Soc. 2025, 147, 46847−46853

Scheme 5. Synthetic Applications



In summary, we have developed a copper-catalyzed asymmetric alkynylation of sp^2 C−H bonds in silyl-substituted allenes. This method provides efficient access to chiral skipped diynes in high yields with excellent enantioselectivities. The method features a broad substrate scope and robust functional group tolerance, accommodating both alkynyl silanes and boronic esters. Moreover, the resulting skipped diynes serve as versatile key synthons for incorporating enantiomerically enriched enynes, alkynes, alkanes, and their derivatives, which could be applied in the concise synthesis of several bioactive molecules.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c19257.

https://doi.org/10.1021/jacs.5c19257
J. Am. Chem. Soc. 2025, 147, 46847−46853

Experimental procedures and characterization data (PDF)

Accession Codes

Deposition Number 2497731 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We are grateful for the financial support provided by the National Key R&D Program of China (2021YFA1500100), the Strategic Priority Research Program of the Chinese Academy of Sciences (No. XDB0610000) and the National Naturale Science Foundation of China (Nos. 22331012, 92256301, and 22171279). P.C. also thanks the Youth Innovation Promotion Association of the Chinese Academy of Sciences (No. Y2022074). G.L. acknowledges support from the Tencent Foundation through the New Cornerstone Science Foundation.

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https://doi.org/10.1021/jacs.5c19257
J. Am. Chem. Soc. 2025, 147, 46847−46853